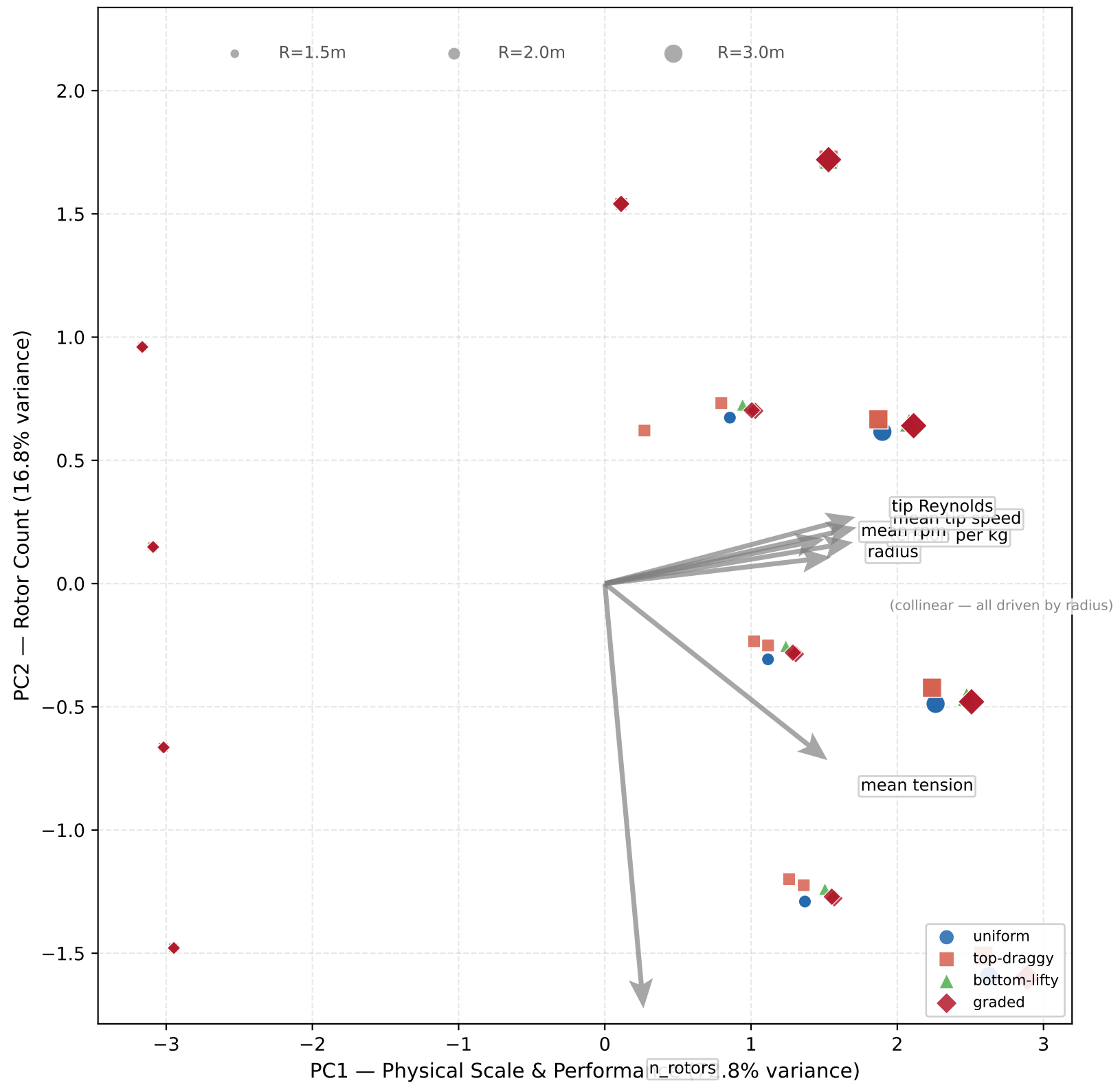


# PCA of BEM Configurations: What Drives Performance Variance?



*IMPLICATIONS: PCA of 96 BEM configurations across 7 variables. The design space collapses to 2-D: PC1 (81%) = physical scale & performance; PC2 (19%) = rotor count. All 4 tilt profiles overlap completely — tilt is a tuning parameter, not a design differentiator. For manufacturing: reduce the 96-configuration space to a 4×4 grid of (radius, N). Build uniform tilt. Data: BEM v2.0, corrected polygon solver. See SPEC.md §6.6.*